# Antimicrobial chemotherapy

By

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## Intended Learning Outcomes (ILOs)

### By the end of this lecture, the student will be able to:

1. **Define** antimicrobial drugs.
2. **Classify** antibiotics (source, action, spectrum).
3. **Differentiate** bacteriostatic vs bactericidal and narrow vs broad spectrum.
4. **Describe** main mechanisms of action.
5. **Relate** antimicrobial action to gene expression.

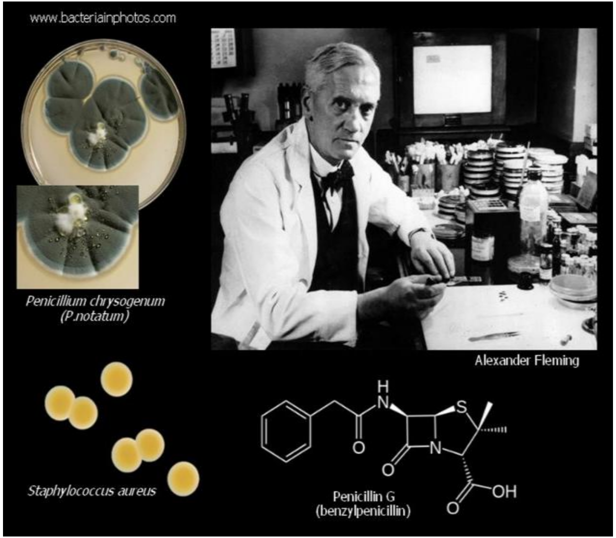
# Antimicrobial Drug

A substance, either synthetic or naturally occurring that kills M.os or inhibits the growth of microorganisms.

# Classifications of antibiotics

## 1) According to the source

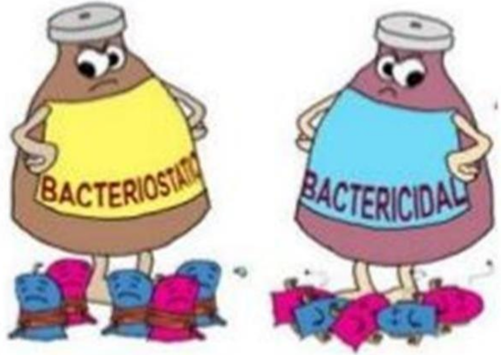
### Naturally




### Synthetic

2) **Based on mode of Action**

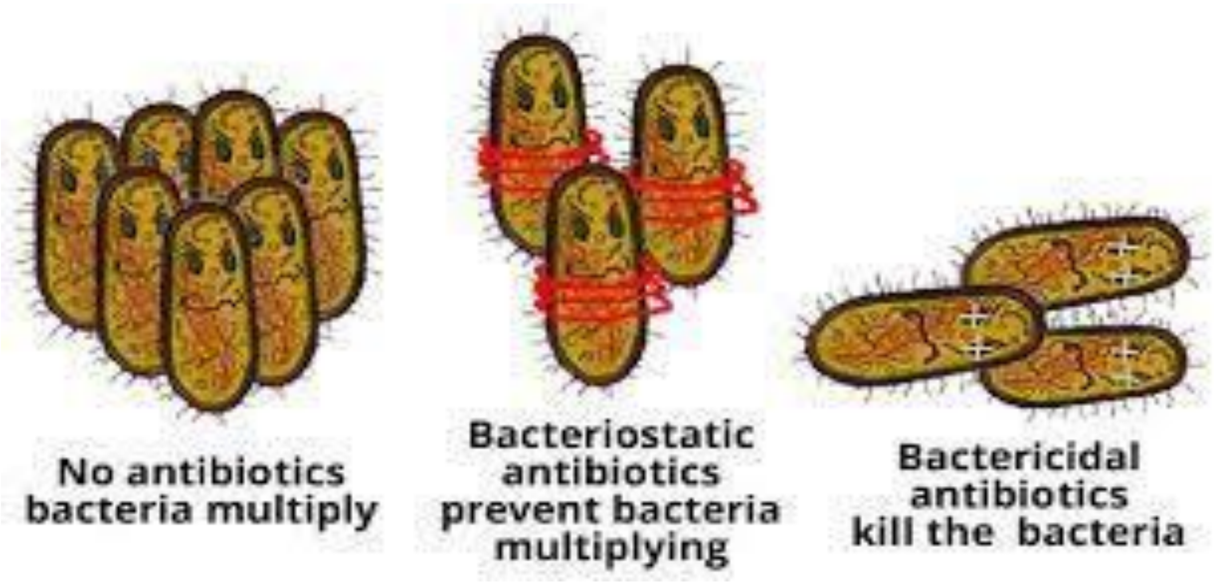
- Bacteriostatic
- Bactericidal



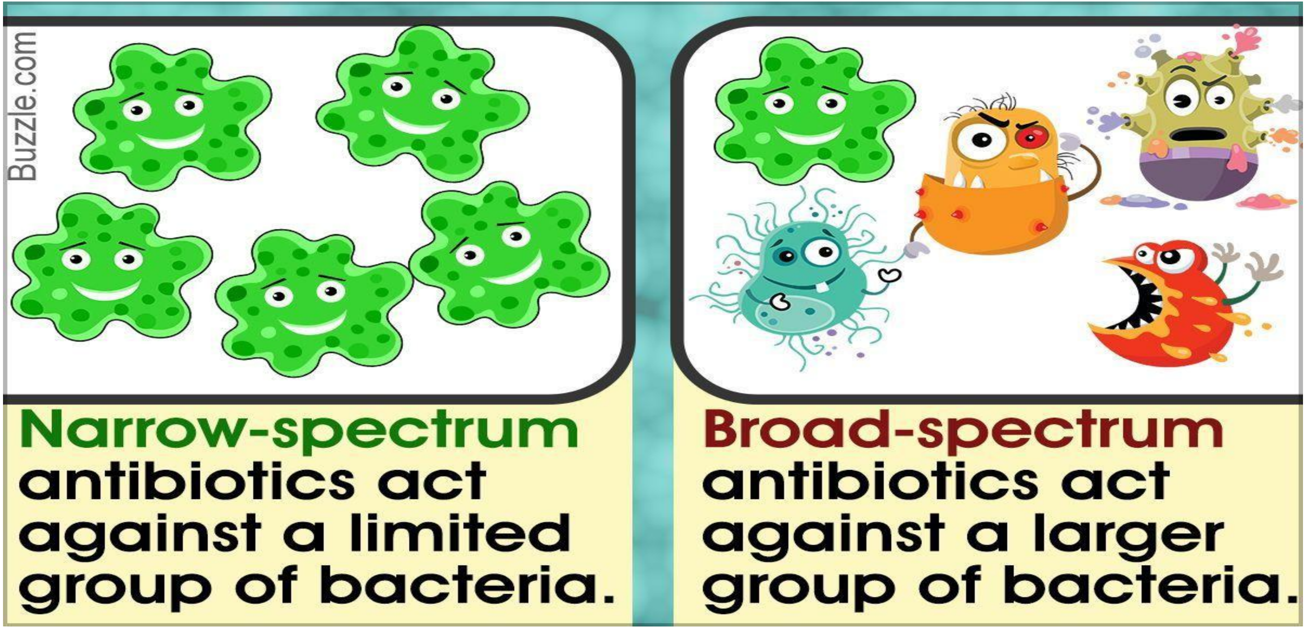

3) **Based on their spectrum of action**

- Broad-spectrum
- Narrow Spectrum

# Bacteriostatic vs Bactericidal



# Narrow spectrum vs Broad spectrum



## 4) According to mechanism of action

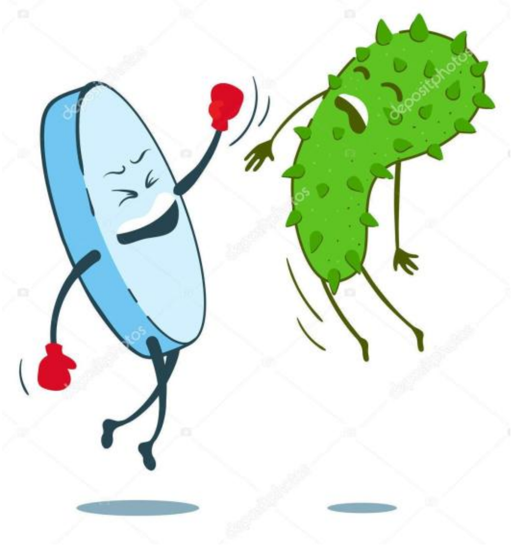
a) Inhibition of cell wall synthesis (Bacterial peptidoglycan)

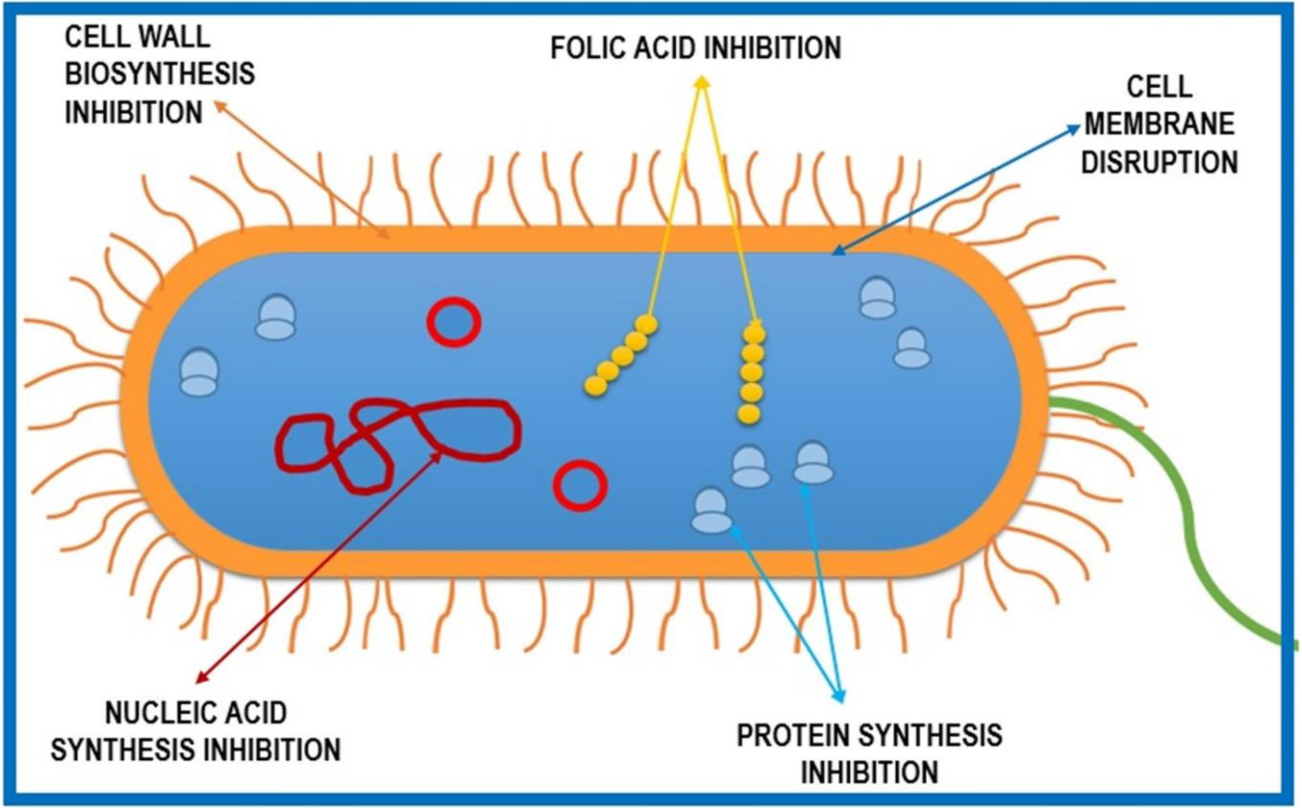
b) Inhibition of cell membrane function.

c) Inhibition of nucleic acid synthesis.

d) Inhibition of protein synthesis.

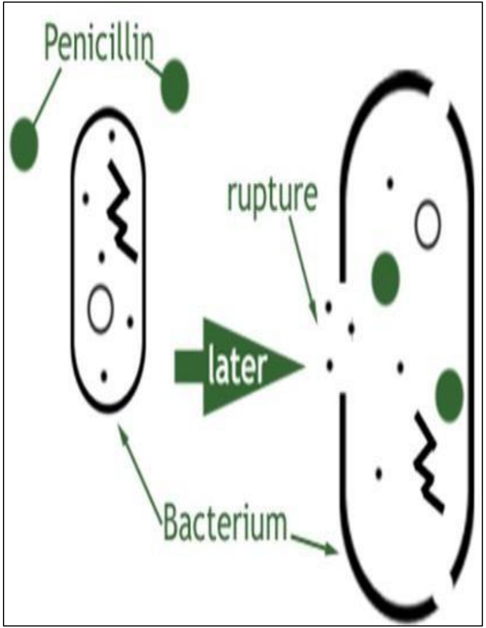
e) Inhibition of metabolic pathway.



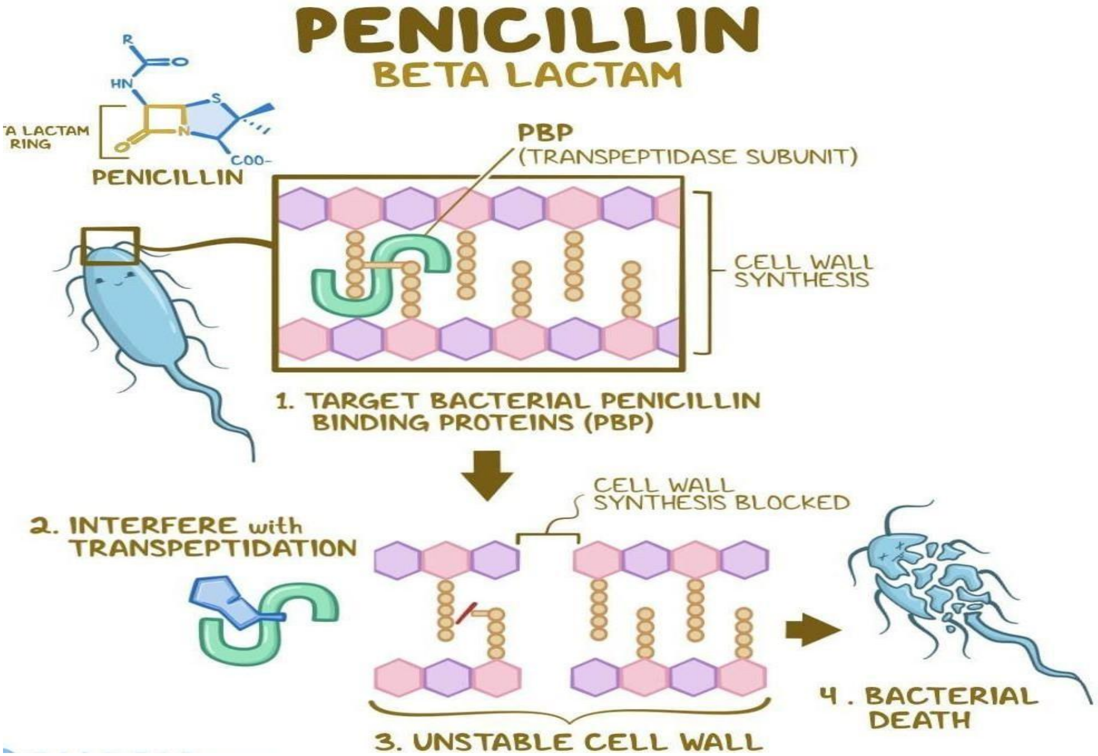

CELL WALL
BIOSYNTHESIS
INHIBITION
FOLIC ACID INHIBITION
CELL
MEMBRANE
DISRUPTION
NUCLEIC ACID
SYNTHESIS INHIBITION
PROTEIN SYNTHESIS
INHIBITION

# 1- Inhibition of cell wall synthesis (Bacterial peptidoglycan):

- Binding of B-lactams to cell wall receptors (PBPs) ⇨ Inhibition of **transpeptidase**, the enzyme that catalyze the **final cross linking step** in the synthesis of **peptidoglycan** ⇨ **cell death and lysis**




- Act only on growing cells (metabolically active cells)
- E.g. **β-lactam antibiotics (Penicillins & Cephalosporins)**



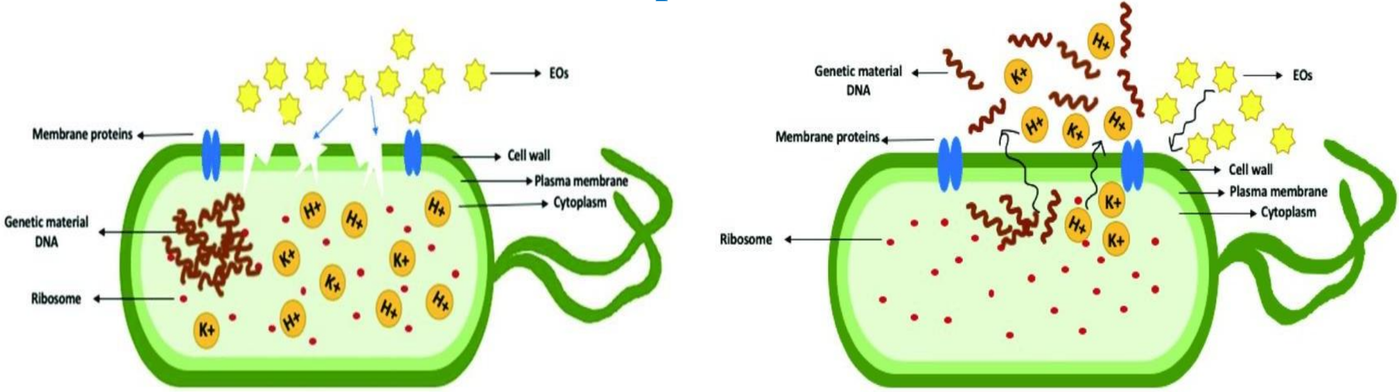

**Bactericidal**

## 2. Inhibition of cell membrane function:

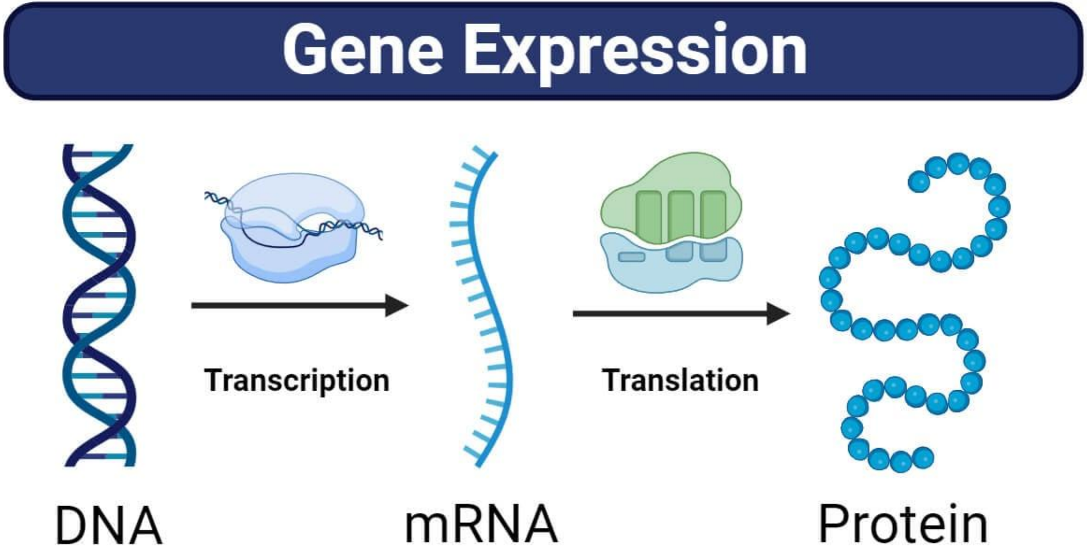
- Change the permeability of the cell membrane ⇨ **leakage of metabolic** substrates essential to life of the microorganism.

- E.g. **Polymyxin B, Amphotericin B**

- **Side effects: neurotoxic and nephrotoxic** “limited in use”



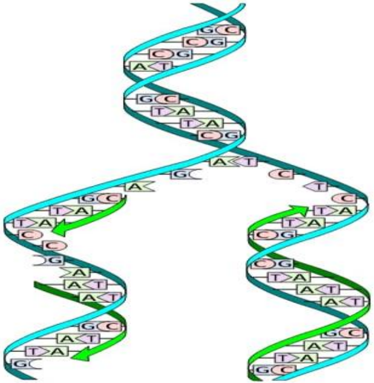
# Gene expression



# 3. Inhibition of nucleic acid synthesis:

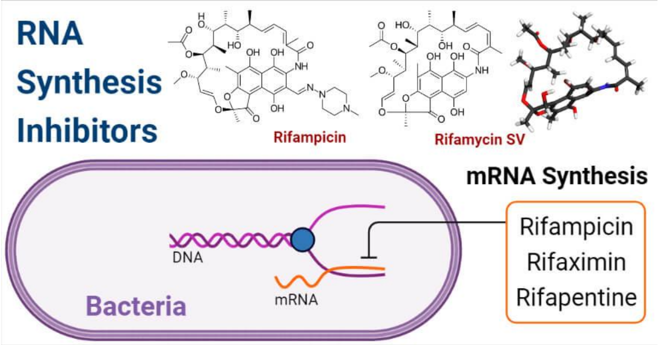
## A. Interference with DNA replication:

- e.g. Quinolones (Ciprofloxacin)



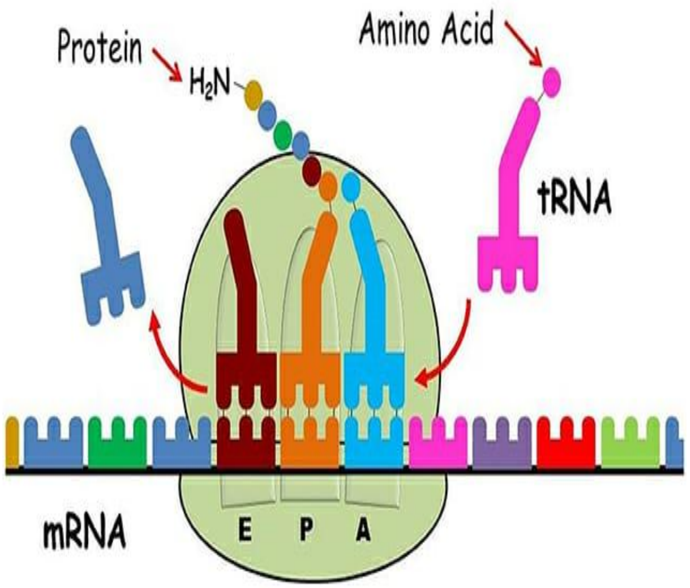
## B. Interference with RNA synthesis:

- e.g. Rifampicin



# 4. Inhibition of protein synthesis:

- Interfere with translation of proteins by the 70S ribosome
- The difference between ribosome of bacteria and mammalian cell (80S) ⇨ not dangerous on protein synthesis of mammalian ribosomes.
- Examples: Aminoglycosides, Tetracycline, Chloramphenicol



**Major classes of protein synthesis–inhibiting antibacterials**

**Chloramphenicol, macrolides, and lincosamides**

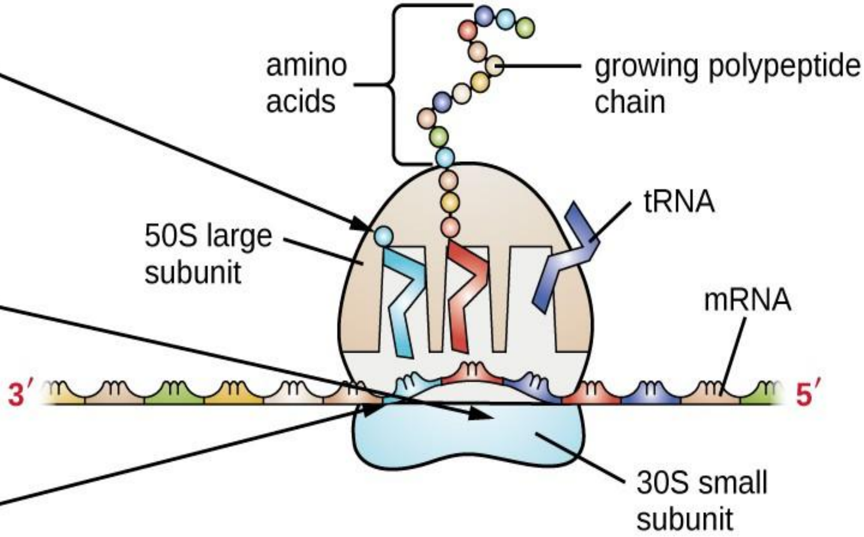
- Bind to the 50S ribosomal subunit
- Prevent peptide bond formation
- Stop protein synthesis

**Aminoglycosides**

- Bind to the 30S ribosomal subunit
- Impair proofreading, resulting in production of faulty proteins

**Tetracyclines**

- Bind to the 30S ribosomal subunit
- Block the binding of tRNAs, thereby inhibiting protein synthesis



30$ **A**minoglycoside

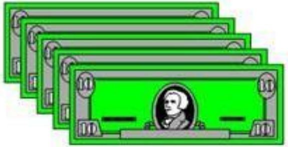
**T**etracycline

50$ **C**loramphenicol

**E**rythromycin

**L**incomycin

c**L**indamycin



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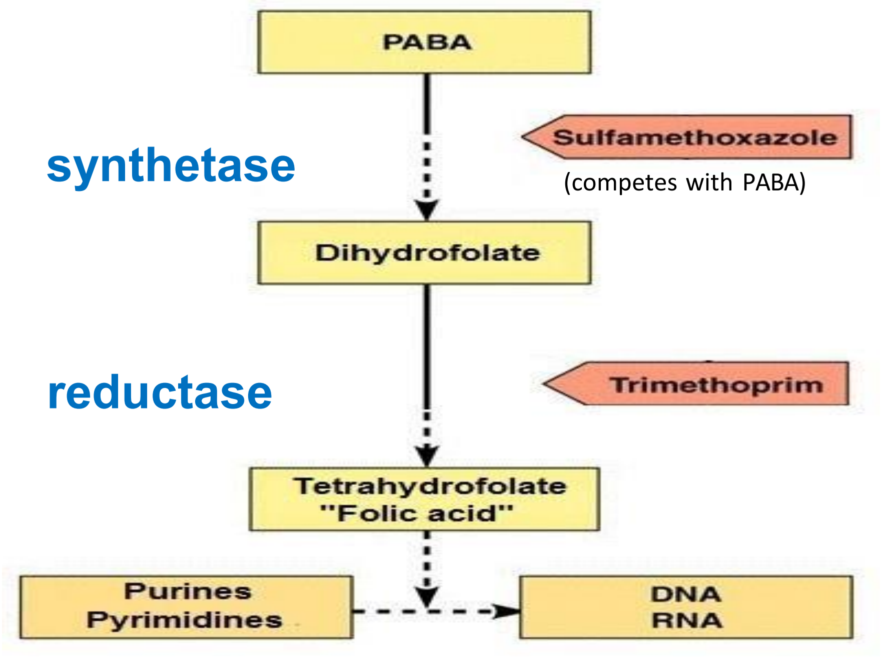
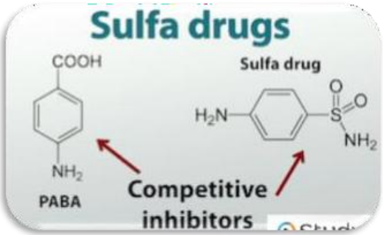
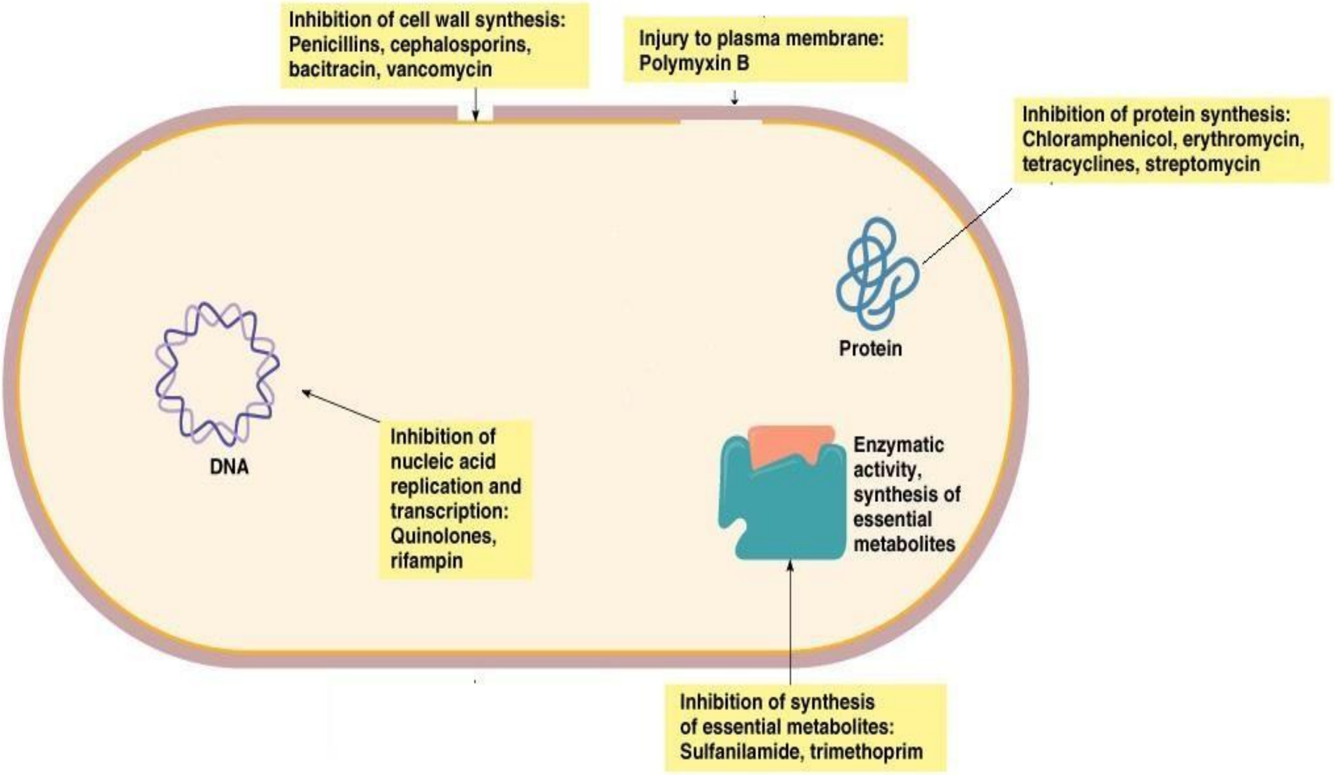
# 5. Interference with essential metabolite synthesis:

## (Folic acid = precursor of purine and pyrimidine bases)




Inhibition of cell wall synthesis:
Penicillins, cephalosporins, bacitracin, vancomycin
Injury to plasma membrane:
Polymyxin B
Inhibition of protein synthesis:
Chloramphenicol, erythromycin, tetracyclines, streptomycin
Protein
DNA
Inhibition of nucleic acid replication and transcription:
Quinolones, rifampin
Enzymatic activity, synthesis of essential metabolites
Inhibition of synthesis of essential metabolites:
Sulfanilamide, trimethoprim


QUESTIONS

**Q1-Bactericidal antibiotics:**

A) Stop bacterial growth

B) Kill bacteria

C) Stimulate immune system

D) Affect viruses only

**Q2-Broad-spectrum antibiotics:**

A) Act on one type of bacteria

B) Act on a limited group

C) Act on a wide range of bacteria

D) Do not affect bacteria

**Q3- Synthetic antibiotics differ from natural ones because they:**

A) Are derived from plants

B) Are produced chemically in labs

C) Are inactive

D) Only affect fungi

**Q4- In immunocompromised patients, bactericidal antibiotics are preferred because they:**

A) Are cheaper

B) Kill bacteria directly

C) Have fewer side effects

D) Act slowly

Q5-The best strategy to reduce antibiotic resistance is:
A) Use antibiotics randomly
B) Use broad-spectrum always
C) Use antibiotics based on culture results
D) Increase dose for all patients

Q6-Overuse of broad-spectrum antibiotics may result in:
A) Faster recovery
B) Increased resistance
C) Reduced infection
D) No effect

# Thank You